**Name:** _____

**Date:** _____

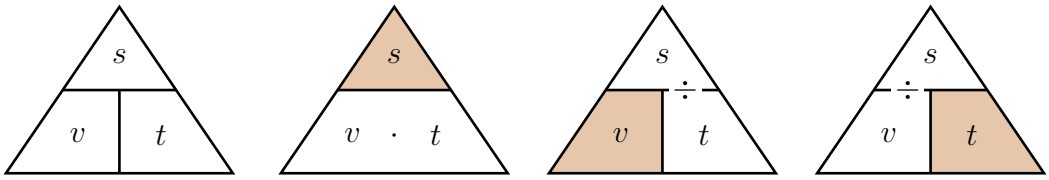
**Mr. Rawson • Y8 MYP Physics**

# Speed Calculations 2: Real Numbers

*Unit 2: Forces and Motion • decimals, units and challenges*

## The triangle, again

Same triangle, same four steps as Sheet 1. Shade the letter you are looking for.




*the triangle* **shade $s$:** $s = v \cdot t$ **shade $v$:** $v = s \div t$ **shade $t$:** $t = s \div v$

## Two new moves

**1. Round to 1 decimal place.** Real numbers rarely divide exactly. Write the calculator answer, then round it: look at the second decimal place, and if it is 5 or more, round up.

**2. Units in, units out.** The answer comes out in the units you put in. Put in kilometres and hours, and a speed comes out in km/h; put in miles and miles per hour, and a time comes out in hours.

**Worked example — copy this shape**

A bus travels **300 m** in **230 s**. What is its speed?

**WRITE** $s = 300$ m $t = 230$ s $v =?$
**CHOOSE** shade $v$: $v = s \div t$
**PUT IN** $v = 300 \div 230 = 1.304\ldots$
**ANSWER** $v =$ **1.3 m/s** (to 1 decimal place)

**Worked example — copy this shape**

A cyclist rides at **20 km/h** for **3 hours**. How far does she ride?

**WRITE** $v = 20$ km/h $t = 3$ h $s =?$
**CHOOSE** shade $s$: $s = v \cdot t$
**PUT IN** $s = 20 \cdot 3$
**ANSWER** $s =$ **60 km** (km/h and hours give kilometres)

**Check:** round once, at the very end, and write the unit after every answer.

## Your turn

Use the four steps for every question. Give speeds and times to 1 decimal place.

**(1)** A student runs **600 m** in **120 s**. How fast is she?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**(2)** A badminton player hits a shuttlecock **50 m** in **2 s**. How fast does it go?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**(3)** A sprinter runs **100 m** in **12.4 s**. What is her speed?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**(4)** A hockey ball is hit at **40 m/s**. How far does it go in **5 s** at that speed?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**(5)** A hockey player runs with the ball at **8 m/s**. How long does it take her to run **80 m**?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**(6)** A plane flies at **350 miles per hour**. London to Singapore is **6750 miles**. How long does the flight take?

**WRITE** _____

**CHOOSE** _____

**PUT IN** _____

**ANSWER** _____

**Challenges — to go further**

**Challenge 1.** Two cars are **4 km** apart, driving the same way along a road. The car behind goes at **10 m/s**; the car in front goes at **9 m/s**. How long until the back car catches up, and how far has each car travelled by then?

*Hint: the gap shrinks by 1 m every second, and 4 km is 4000 m.*

**Challenge 2.** (a) What is **100 km/h** in m/s? (b) What is **100 miles per hour** in m/s?

*Hint: 1 km = 1000 m, 1 mile = 1609 m, and 1 hour = 3600 s. Change the distance to metres and the time to seconds, then divide.*

**Challenge 3.** Two cars are **20 km** apart, driving towards each other. One goes at **40 km/h** and the other at **80 km/h**. (a) How long until they meet? (b) Where do they meet?

*Hint: together they close the gap by* $40 + 80 = 120$ *km every hour.*